Pranjal Kalekar

Boston | pranjalkalekar10@gmail.com | pranjalkalekar.me

SUMMARY

Applied Data Scientist with a proven history of spearheading cross-functional data initiatives from initial prototype to global scale. Dedicated to delivering secure, high-quality AI and data products that drive measurable growth and operational excellence.

TECHNICAL SKILLS

- **Tech Languages**: Python, MySQL, MS SQL, R, MATLAB, No-SQL, C++, HTML/CSS
- **Databases**: MongoDB, MariaDB, Neo4j, Apache Spark, PostgreSQL, NoSQL, RDBMS
- **Tools**: Git, VS code, PyCharm, Google Cloud Platform, Tableau, PowerBI, WordPress, Azure, Hadoop
- **Libraries**: PyTorch, Plotly, Tensorflow, SciPy, pandas, Scikit-learn, dlib, seaborn, NLTK, XGBoost, NetworkX
- **Platforms**: Microsoft Excel, Google Colab, PySpark, MS Office, Unix/Linux, MySQLWorkbench, R Studio

EDUCATION

Northeastern University — **Sep 2022 - Dec 2024**
Master of Science, Data Analytics Engineering — *Boston, MA*

VJTI — **Aug 2018 - May 2022**
B.Tech, Electronics Engineering — *Mumbai, MH, India*

EXPERIENCE

Haas Lab, Mass General Hospital- Harvard Medical School
Research Engineer — *Boston, MA* · **Mar 2025 - Present**

- Implement data preprocessing steps, baseline models, and evaluation approaches for graph neural network models.
- Stabilize transformer-based GNN training by refactoring the architecture and applying systematic hyperparameter tuning in PyTorch, surpassing benchmark performance by 15% and improving model reliability. Deploy GPU-optimized stacking of XGBoost and neural networks, enhancing correlation by 14% over baseline models and isolating critical biological feature importance.
- Engineered a graph-based imputation pipeline using machine learning, reducing data sparsity and achieving 35% better performance while applying responsible AI practices with Copilot and Claude Code.
- Optimized GPU-accelerated workflows to execute Non-Negative Matrix Factorization (NMF) pipelines for decomposing and reconstructing complex proteomics, including single cell liver datasets, as well as contributing to active grant proposals.

Data Analyst-1 — *Boston, MA* · **Jan 2024 - Sep 2024**

- Analyzed breast cancer proteomics data from mass spectrometers using clustering methods including K-means, hierarchical clustering, and DBSCAN; dimensionality reduction methods including UMAP, PCA, and t-SNE; and imputation methods including Combat and LUPINE to improve data quality and clarify tumor subtypes.
- Trained a node-classification model that achieved a maximum 96% AUC-ROC for individual graphs; created posters and presented findings at research gatherings to support collaborations.

DataZ
Data Associate — *Remote - Boston, MA* · **Jan 2025 - Mar 2025**

- Built Python-based web scraping pipelines using Selenium, Scrapy, BeautifulSoup, and ScraperAPI to collect product data from sites like Alibaba and AliExpress, and stored 30K+ entries in a MySQL database, enabling structured analytics and automated reporting
- Delivered actionable insights and Tableau dashboards tracking key engagement metrics, supporting investor updates and product decisions in a lean early startup

PROJECTS

Fuel-IT-Right — **Nov 2025**
Sundai

- Live Calorie burn calculator for different intensity level workouts using different sensors on an athlete
- Encapsulated in a apple watch app and a webpage connected to heart rate and power sensors using Swift Studio

Satellite Imagery Gap Filling using CNN — **Sep 2025**
Wood-well Climate research center

- Build a CNN Model to predict the gaps in the Landsat dataset for further processing, trained onthe AlphaEarth and modis dataset
- Ran ETL process using Python on LendingClub data for extraction, cleaned, processed, and fine-tuned the data for future extractions with correlation with the target

LLM based OS agent for iOS — **May 2022**
Sundai

- Developed a stateless AI Agent orchestration layer that integrates LLMs (GPT-4o) with native macOS UI automation to perform real-time task execution
- Reduced pipeline latency by optimizing core script execution and implementing efficient data handling techniques
- Leveraged native macOS Accessibility permissions (AXIsProcessTrusted), ensuring full compliance with system-level security protocols and explicit user consent

CERTIFICATES AND LEADERSHIP

Kaggle Advanced SQL certificate, Graduate Certification in Leadership with Northeastern University, Visualizing Land Cover and Land Use Change with NASA Satellite Imagery, Monitoring Water Quality in Lakes and Coastal Regions Using STREAM(ARSET)

Hacker at Sundai MIT- Harvard AI Prototype Launching Club, Kaggle Advanced SQL certificate, Former VP of Relations and Projects Data Science Hub, Graduate Certification in Leadership with Northeastern University, MIT- ESP splash Teacher 2025, NU cycling- Road Biking ECCC Racing